

Full VAPT Cycle Report

Executive Summary

A Vulnerability Assessment and Penetration Testing (VAPT) exercise was conducted against the Metasploitable2 virtual machine. The assessment included reconnaissance, vulnerability identification, exploitation, web application testing, and remediation recommendations.

Assessment Methodology

1. Information Gathering
2. Vulnerability Identification
3. Exploitation
4. Post-Exploitation Analysis
5. Reporting

Target Information

Target IP:
192.168.56.101

Environment:
VirtualBox Lab Environment

Discovery Phase

Tool Used:
Nmap

Command:
nmap -sV 192.168.56.101

Results:
Multiple vulnerable and outdated services were identified, including FTP, Samba, Apache, Tomcat, MySQL, PostgreSQL, and UnrealIRCd.

Vulnerability Assessment

Key Vulnerabilities:

- Outdated FTP Service
- Exposed Tomcat Manager Interface
- Weak Service Configurations
- Web Application Vulnerabilities

Exploitation Phase

The Tomcat service running on port 8180 was identified and administrative interfaces were accessed during testing.

Risk Level:

High

Web Application Testing

The DVWA platform was used to validate:

- SQL Injection
- Cross-Site Scripting
- Request Manipulation

Tools:

- Burp Suite
- OWASP ZAP

Recommendations

1. Apply security patches regularly.
2. Disable unused services.
3. Restrict management interfaces.
4. Implement secure coding practices.
5. Perform periodic vulnerability assessments.

Conclusion

The assessment successfully demonstrated the complete VAPT lifecycle. Multiple security weaknesses were identified and analyzed. Remediation measures have been provided to reduce organizational risk and improve the overall security posture.

Non-Technical Management Brief

A security assessment was conducted on a test environment to identify weaknesses that could be exploited by attackers. Several outdated services and vulnerable web applications were discovered. While the environment was intentionally insecure for training purposes, the findings demonstrate common risks found in real-world systems. Implementing security updates, strengthening access controls, and performing regular assessments will significantly reduce the likelihood of compromise and improve overall security resilience.